

SUMMARY

CS & AI undergrad building production-grade full-stack systems and AI-powered applications, from LLM agent frameworks and ML pipelines to deployed REST APIs and cloud-native infrastructure.

EDUCATION

Newton School of Technology, Rishihood University2023 – 2027

B.Tech in Computer Science & AICGPA: 7.6/10

EXPERIENCE

- Frontend Developer InternMay 2025 – Nov 2025

Rishihood UniversitySonipat, Haryana

 - Built a **Fee Payment Module** enabling seamless online payments for university students with secure transaction flow and state-managed UI using **Next.js**, **React.js**, and **Redux Toolkit**.
 - Developed a **Learners Management System** used by **3000+** students and admins, with optimized data handling, role-based access, and engagement-focused UI.
- Frontend Developer InternMar 2025 – Apr 2025

Wonder Creative StudioRemote

 - Developed frontend pages for Automation and Web-App modules using **Next.js**, **React**, **Node.js**, and **Express.js**; contributed full-stack features including **JWT**-based authentication and dashboard functionality.
- Tech & Business Growth InternJan 2025 – Feb 2025

PlusCashBengaluru, Karnataka

 - Designed database models for mutual funds, indices, and factsheet data; integrated APIs and automated financial data ingestion workflows using **React.js**, **TypeScript**, **SQLite**, and **AWS S3**.
- Frontend Developer InternJun 2024 – Jul 2024

BadBusiness.InRemote

 - Built dynamic book sections page and contact form using **HTML**, **CSS**, **JavaScript**, and **emailJS**; optimized site performance by replacing heavy image assets with lightweight links.

PROJECTS

- AI Agent Orchestration Framework | TypeScript, Next.js, Gemini API2025

 - Built a production-grade **multi-step AI agent framework** integrating **Google Gemini** with structured plan generation, **Zod** schema validation, and LLM-based semantic verification using **TypeScript** and **Next.js**.
 - Implemented reliability patterns including **circuit breaker**, exponential backoff retry, **DAG-based parallel execution**, partial replanning, and a memory layer tracking tool reliability and failure patterns across iterations.
- SnackyNerds: Full-Stack App with CI/CD Pipeline | React, Node.js, Docker, Terraform, AWS2025

 - Designed a **5-phase CI/CD pipeline** using **GitHub Actions** automating testing (Jest, Vitest, Playwright), infrastructure provisioning, and dual deployment to **ECS Fargate** and **EKS**.
 - Provisioned AWS infrastructure (**ECR**, **ECS**, **EKS**, **S3**, **IAM**, **VPC**) via **Terraform IaC**; deployed with **2-replica** Kubernetes setup with liveness/readiness probes and resource limits.
- Adaptive Cyber-Physical Security | TensorFlow, scikit-learn, Python2025

 - Developed a hybrid ML-based IDS trained on CIC-IDS-2018 with **8.2M+** network flows and **13** attack families; achieved **ROC-AUC 0.8867** and precision **0.8179** using Denoising Autoencoders.
 - Implemented **Random Forest**, One-Class SVM, Autoencoder, Isolation Forest, and ensemble detection pipelines for zero-day threat generalization.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C, Bash
AI/ML: TensorFlow, scikit-learn, LLM Integration (Gemini, Claude), Pandas, Zod, Agent Frameworks
Frontend: React.js, Next.js, Redux Toolkit, Tailwind CSS, Framer Motion
Backend/Infra: Node.js, Express.js, MongoDB, Prisma ORM, Docker, AWS, Kubernetes, GitHub Actions, Terraform

LEADERSHIP & ACTIVITIES

- Founder and Lead of Society for Aerospace and Space Technology at Rishihood University.
- Configured **4 Intel i7 Linux servers** with Ubuntu 24.04/Fedora, including DHCP, NAT, Kubernetes (from scratch), containerd (CRI), CNI, and CSI integrations; guided students in hands-on systems learning.
- Placement Committee Representative for **103 students**; Open Source Contributor at GirlScript Summer of Code and Global Open Source Cohort.

CERTIFICATIONS & ACHIEVEMENTS

TryHackMe Pre Security Certification | Cyfrin Updraft Blockchain Basics | OverTheWire Bandit Completed
TryHackMe: astroroot6 (**177K+** points, Hacker Level 8) | Hands-on LLM Jailbreaking using GraySwan Arena